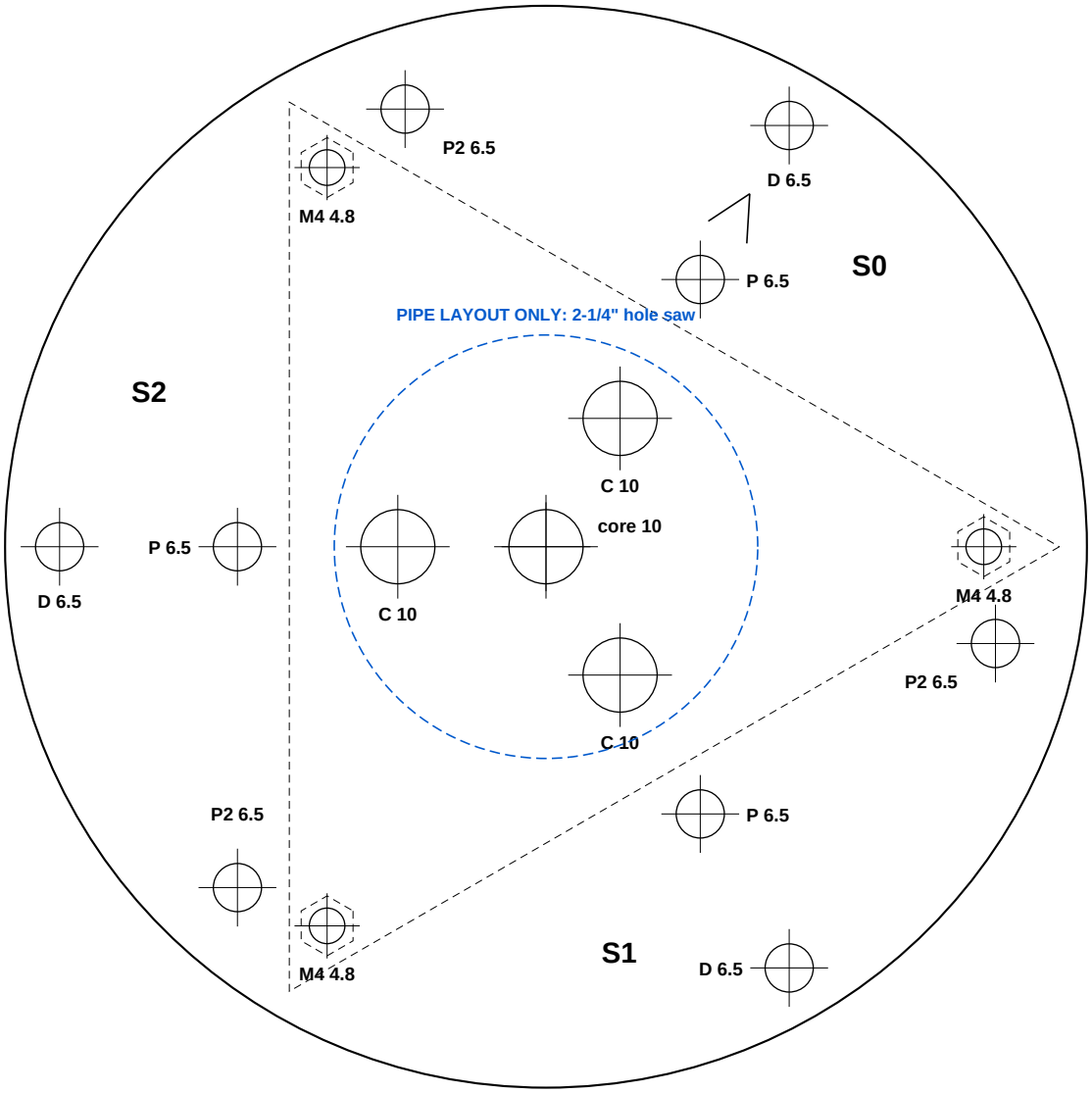
Shield deck drill template - 146 mm disc - PRINT AT 100% (Actual size)

Viewed from the TOP (electronics side). Tape to the board, centre-punch each crosshair, then drill.



Solid circle = cut line (146 mm). Dashed triangle = where the 120 mm back panels stand, copper side out.

D 6.5 mm = rubber duck / monopole / dipole whip (31 mm in front of face) P / P2 6.5 mm = patch (7 mm in front of face; P2 14 mm from the face end)

C 10 mm = dipole coax pass-through (inside core) M4 4.8 mm = bolts to the plate, nut flush in each core corner

Core 10 mm at the centre = antenna coax into the core; open it up with a hole saw (centre mark) if needed.

Blue dashed 57.1 mm (2-1/4" hole saw) = pipe layout only (page 4): the pipe passes into the core;
it removes the C holes. Other layouts keep the 10 mm core hole.

Arrow points to Sector 0. Holes are symmetric, so the template also works flipped (labels read mirrored).

Drawn the same way round as the plates: plate 12 o'clock up, S0 arrow 30 deg clockwise of it.

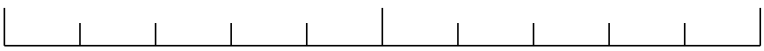
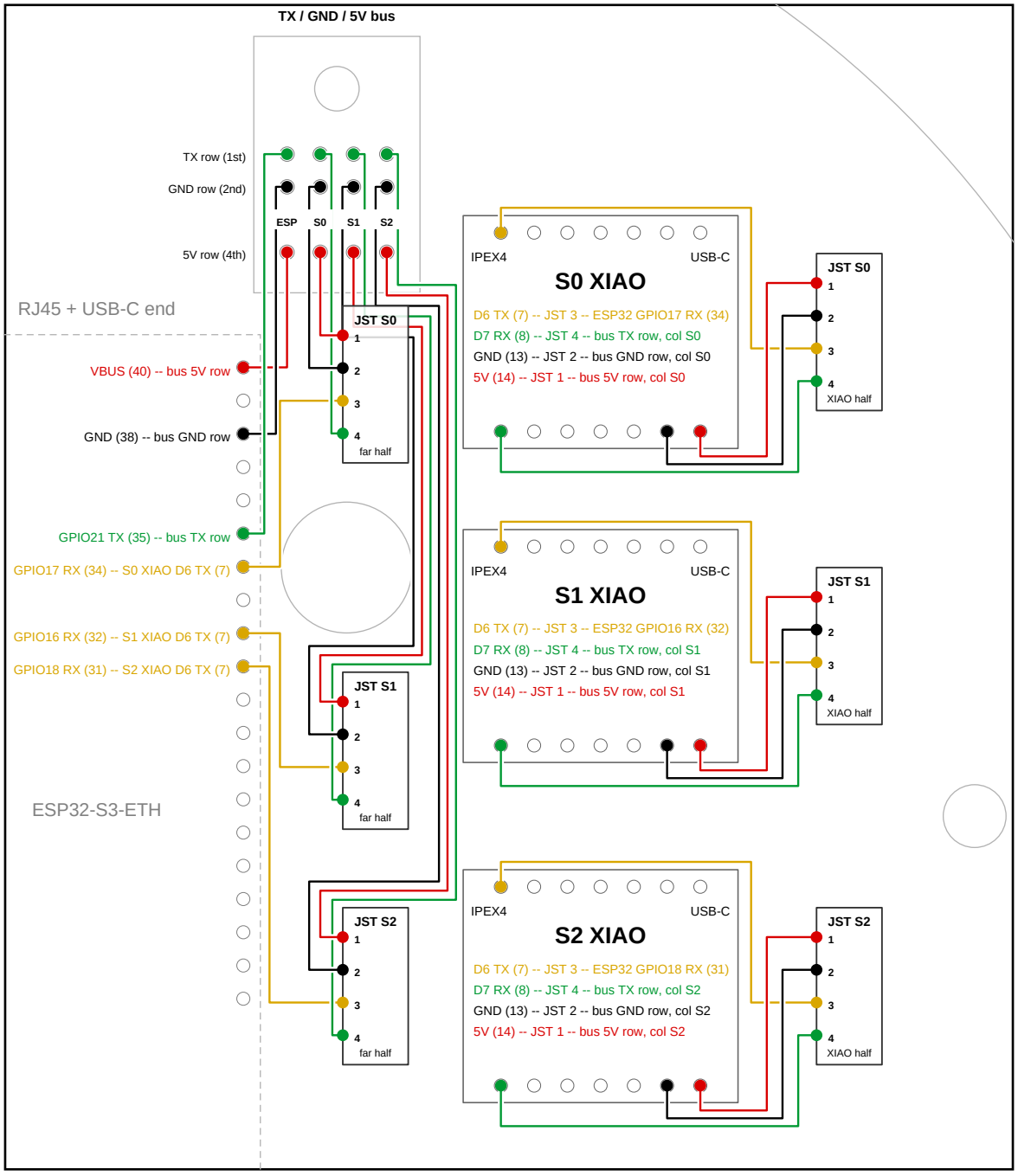


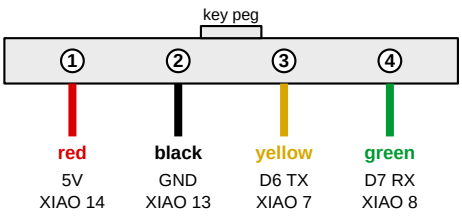
Plate wiring reference - original plate layout (its drill template is retired)

Top view of the radio side, 2x (not for drilling). Pin numbers in brackets. Pad positions approximate.
JST halves drawn apart: XIAO half right of each radio, far half in the strip; same name = they plug together.



D6 -> its own ESP RX (the pin is the sector): S0 GPIO17 (34), S1 GPIO16 (32), S2 GPIO18 (31).
ESP TX GPIO21 (35) -> bus TX row -> every D7. Feed: TX (35) -> TX row, VBUS (40) -> 5V row, GND (38) -> GND row.
Bus: hole end up, rows bridged, 3rd row spare. ESP32 RJ45 side up, RJ45 / USB-C end north: VBUS (40) top right pin.

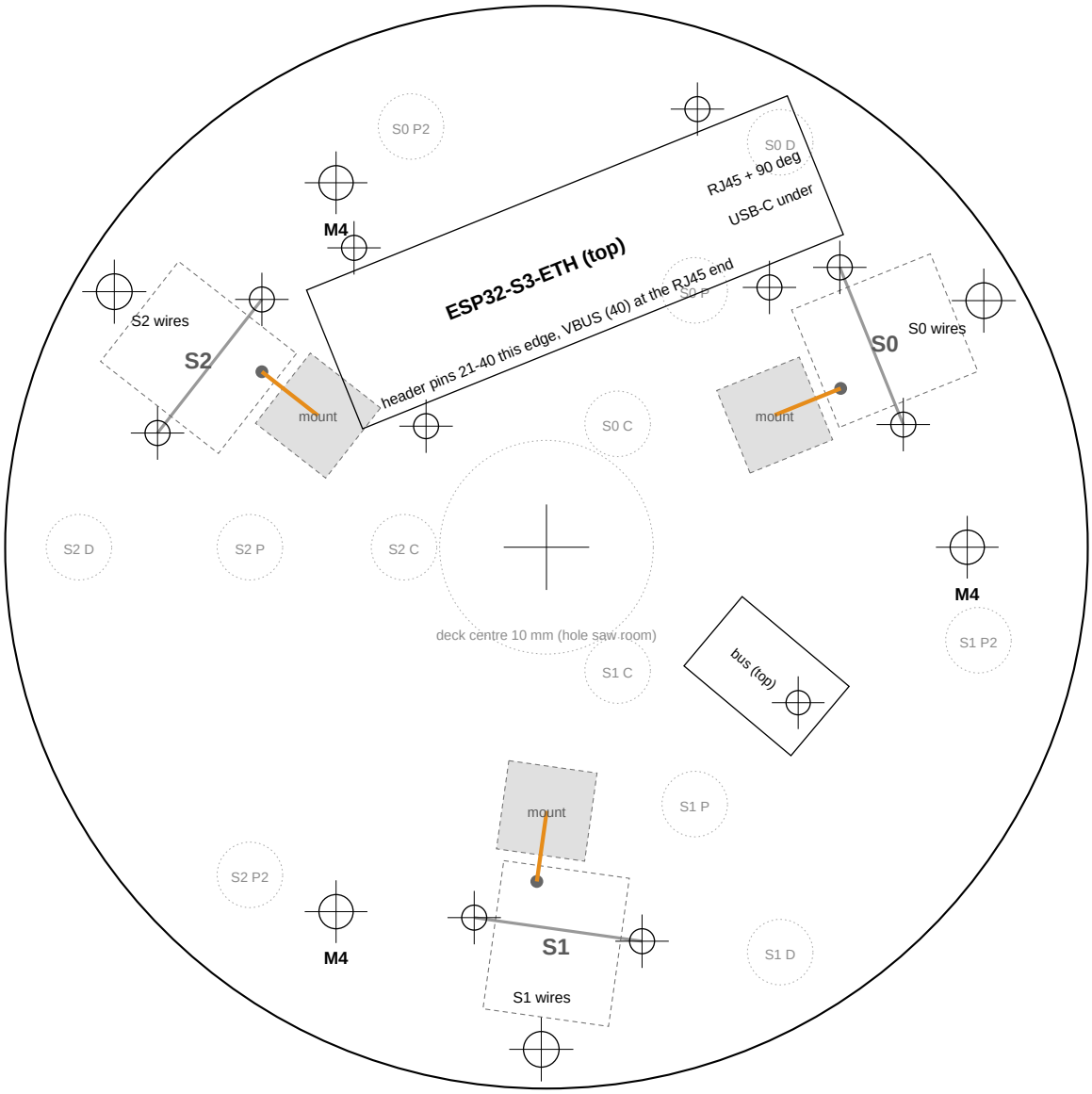
JST head (either half), crimp side, key peg up:



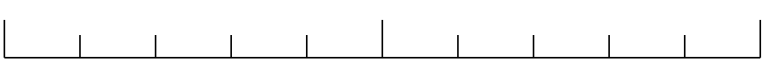
Pins left to right: 1 red 5V, 2 black GND, 3 yellow D6 TX, 4 green D7 RX.
Male and female halves carry the same colours in the same order on the crimp side, so the XIAO tail and the far tail match colour for colour through the joint.
XIAO tail: red -> 5V (14), black -> GND (13), yellow -> D6 (7), green -> D7 (8).
Far tail: red -> bus 5V row, black -> bus GND row, green -> bus TX row, yellow -> that sector's ESP RX pin (labels on the drawing). Female half on the far tail keeps 5V covered.

ALT electronics plate (split) - 152 mm disc - PRINT AT 100% (Actual size)

Viewed from the TOP. Radios on the UNDERSIDE (grey, seen through the plate); ESP32 + bus on top. No pigtail hole.

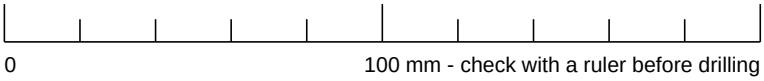


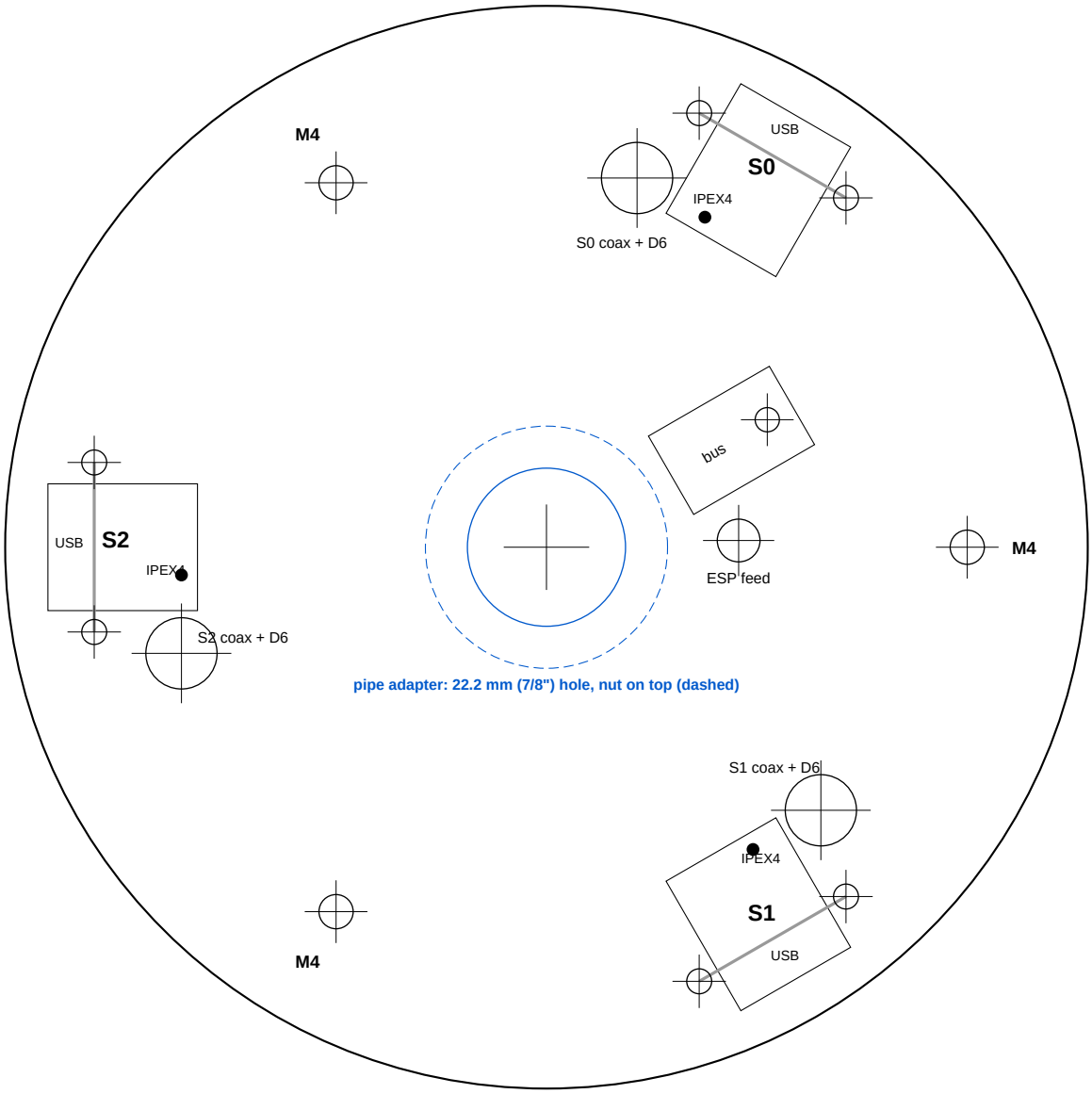
Drill: 3 x 4.8 mm (M4) bolts to the deck, on the deck's M4 holes; 3.4 mm (M3) bus board; 3 x 5 mm harness holes (chamfered): each XIAO's 4-wire tail up to its JST pair on top; 3.5 mm pairs: one tie across each XIAO, two across the ESP32. Underside (grey): XIAOs on VHB + a zip tie (grey bar), IPEX4 dot -> orange flat coax run -> HS-100 tie mount, which clamps it; bend only after the mount, down to its deck jack. Dotted circles: deck jacks below, not drilled. Top: ESP32 parallel to S0, RJ45 / USB-C end toward 2 o'clock. A 90 deg adapter turns the RJ45 up; patch + USB-C lead run through the 4" box's 1" hole to the PoE splitter. USB-C is under the board: slim plug. Spacers 25-30 mm. S1 / S2 are S0 turned 120 deg about the centre. Underside keeps clear of the deck's centre hole (30 mm, assumed). OPTIONS (builder's choice): this a



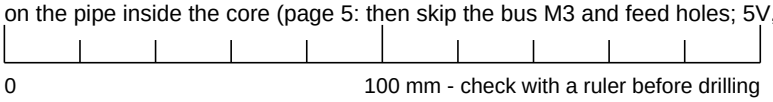
PIPE electronics plate - 152 mm disc - PRINT AT 100% (Actual size)

Viewed from the TOP. XIAOs + bus on top, pipe adapter through the centre; PoE splitter + ESP32 inside the core.



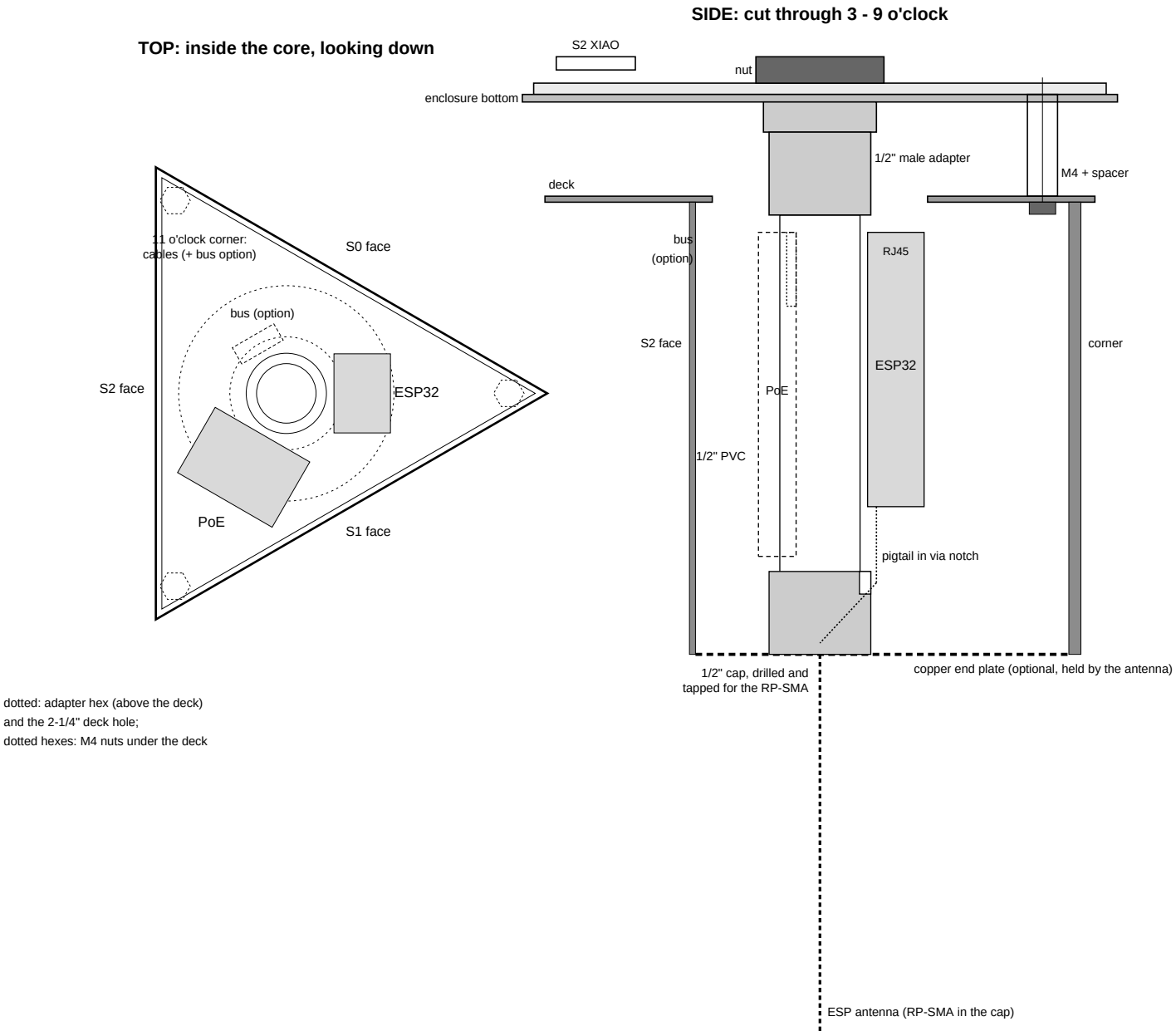


Drill: 3 x 4.8 mm (M4) bolts to the deck, on the deck's M4 holes; 3.4 mm (M3) bus board; 3 x 10 mm pigtail holes;
6 mm ESP feed hole (TX, GND, VBUS up from the core to the bus); 3.5 mm pairs: one tie across each XIAO.
Centre (blue): 22.2 mm (7/8") hole for the 1/2" male adapter's thread, which also goes through the
enclosure bottom; the nut on top (dashed, size est.) clamps both. The PoE cable passes the plate rim (pipe is capped).
Each XIAO sits over its own sector: pigtail up its hole to IPEX4, its D6 wire back down the same hole to its
ESP32 RX pin; 5V, GND and D7 over the top to the bus. IPEX4 (dot) is on the right seen from the top with
USB-C toward you, so each pigtail hole is on that side. USB-C toward the rim, clear of the M4 bolts.
OPTIONS (builder's choice): this pipe plate or the alt split plate (page 3); bus board here beside the nut, or
on the pipe inside the core (page 5: then skip the bus M3 and feed holes; 5V, GND, D7 go down the pigtail holes).



Pipe stack - top and side views - scale 0.6 (NOT a drilling template)

Indoor way up: electronics plate on top, core and antennas below. Same plate frame as page 4 (12 o'clock up).



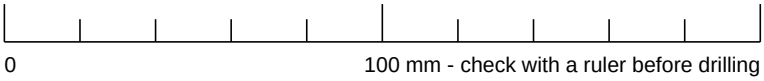
dotted: adapter hex (above the deck)
and the 2-1/4" deck hole;
dotted hexes: M4 nuts under the deck

Pipe: 1/2" PVC (21.3 mm OD) from the male adapter's socket through the 2-1/4" deck hole, ~135 mm long (check the socket depth), capped flush with the core's far end. The cap is drilled and tapped: the ESP antenna's RP-SMA bulkhead threads into it, and the U.FL pigtail from the ESP32 enters through a notch in the cap rim. The antenna, screwed onto the bulkhead, clamps the copper end plate that closes the core (optional; shape not final, must leave access).
Splitter and ESP32 zip-tied to the pipe facing the 3 and 7 o'clock corners (widest room), RJ45 / USB-C end up; the 11 o'clock corner is left for cables (and the bus option). PoE cable: down past the plate rim, through the deck hole.
The adapter's thread goes up through the enclosure bottom and the plate; the nut on top clamps both.
OPTION: the bus board can go on the pipe (dashed, 11 o'clock, opposite the ESP32 and splitter) instead of on the plate.
Side: dashed box = PoE splitter, behind the cut. XIAOs, pigtail holes: as page 4. Estimates: adapter, nut, enclosure bottom, splitter size, board depths, antenna length.

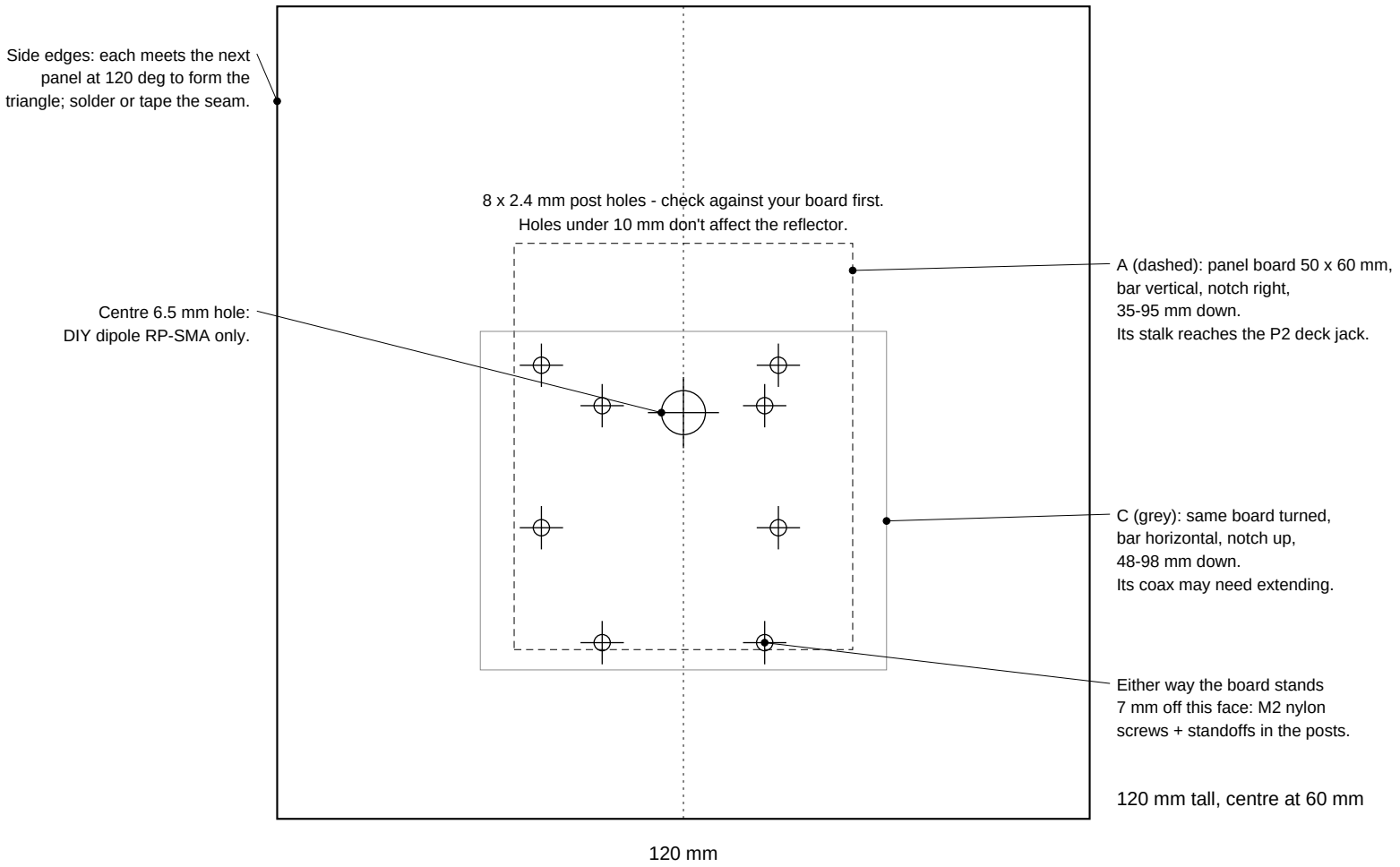
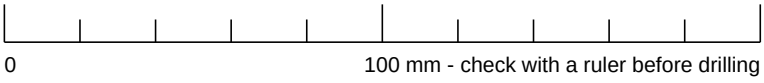
50 mm at drawing scale

Back panel (reflector face) template - 120 x 120 mm - PRINT AT 100%

Make 3. Viewed from the FRONT (copper side, facing the antenna). Top edge goes against the shield deck.



TOP - against shield deck



Whips (rubber duck / monopole / dipole) don't mount on this face: they hang on the deck's D holes, 31 mm in front of it (page 1).